21	$\frac{1}{3}$	P1	for showing $x^2 + y^2 = \frac{13}{36}$ oe	for showing $\left(\frac{x}{n}\right)^2 + \left(\frac{y}{n}\right)^2 = \frac{13}{36}$ oe	Assuming x =P(R) and y =P(Y)
		P1	for setting up an equation in x eg $(1 - x)x = \frac{1}{4}$ oe	for using $x^2 + y^2 = 13$	
		P1	for a process to solve the equation in x	for showing eg $\frac{y}{n}\left(1 - \frac{y}{n}\right) = \frac{1}{4}$ oe	
		P1	for a full method to find P(Y)		
		A1	$\frac{1}{3}$ from correct working		An answer of $\frac{1}{3}$ alone is insufficient for any marks; it must be supported by working shown.

Documented scheme clarification

1MA1_MOCK_SET_04_2H_Q21

Original source

scheme page 12, SHA256
1b50aa06783c6cc171173cc3d7916bc8bad688a9b8e3c7e4c89ccac00764a3a8

Original wording

Some alternative additional guidance drops the denominator36 or uses a yellow variable for the red condition.

Correction / interpretation

Use r=P(red), y=P(yellow). The given conditions are r^2+y^2=13/36 and r(1-r)=1/4. Thus r=1/2 and y=1/3. Some alternative additional-guidance expressions omit /36 or apply the red condition to the wrong variable; the main answer 1/3 is correct.

Reason

Replacement gives independent draws; solving the two actual conditions fixes the stated answer.

Sol authority

6556bd9484cf36b0c354ee080554a4708aede7b6e7124c68cbeb9eaa1e057175